

A Structural Evidence Gate for Multi-Agent Claim Verification: Verbatim-Grounded Agreement, and When It Helps

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Abstract

Background. Multi-agent large language model (LLM) pipelines verify claims by letting model instances debate or critique each other, but such opinion-only exchange can fail by *agreeing on the wrong answer*: nothing in it structurally prevents correlated hallucination. **Objective.** We ask whether constraining agreement *structurally* — permitting an affirmative or contradictory verdict only when it quotes a verbatim span of retrieved evidence, checked programmatically — catches more hallucinations than a single LLM, and at what accuracy, latency and cost, across model scale and domain. **Method.** We build Aletheia, a deployed LangGraph pipeline whose verifier is downgraded to *Unverifiable* whenever it cannot produce an exact evidence substring, and evaluate it against a single-LLM baseline and an ungrounded multi-agent ablation on SciFact and FEVER, grounding all arms in one frozen, versioned corpus with paired significance testing. **Results.** On the live model (Gemini flash-lite, n=100) the grounded arm ties the baseline on accuracy (79.0% vs 79.0%) while improving catch rate (96.6% vs 93.1%) and false-agreement (6.1% vs 10.5%) directionally; on a weaker 8B model the catch-rate gain is large and significant (82.8% vs 60.3%, delta +22.4 pp, 95% CI [+12.1, +33.3]). A cross-model sweep (8B–550B) shows grounding’s accuracy contribution is *inversely related to base-model strength*. On FEVER a corpus-cleaning fix recovers +21 pp of grounded accuracy (56.0%→77.0%, McNemar $p = 4.9e-5$), and the load-bearing guarantee holds exactly: of 51 asserted verdicts, 51/51 (100%) are backed by a verbatim-verified span. **Conclusion.** A structural evidence gate is a mechanically verifiable defence against false agreement whose catch-rate direction holds across three scales and two domains, and whose accuracy effect is a genuine, scale- and domain-dependent trade-off rather than a free lunch.

Keywords: automated claim verification; large language models; hallucination detection; multi-agent systems; retrieval-augmented generation; evidence grounding; fact-checking; reproducible evaluation

1. Introduction

The reliability of large language models (LLMs) is limited by *hallucination* — fluent output unsupported by any source. The problem is most costly in **automated fact and claim verification**, where the failure mode is not an obvious error but a *confident, wrong verdict*: a system that labels a false claim “supported,” or asserts more than its evidence allows, actively manufactures misinformation rather than merely failing to catch it.

A structured review of 35 empirical journal studies on LLM factuality, hallucination, and verification (summarised in the supplementary literature table) reveals a field converging on partial solutions with one shared blind spot. Two problem framings dominate: **factual-accuracy benchmarking** — does the model get the answer right (Yanagita 2023 report GPT-4 at 81.5% on a medical licensing exam) — and **hallucination detection** — is the output unsupported (Massenon 2025 taxonomise real user-reported hallucinations; a clinical-safety framework measures a 1.47% hallucination rate in medical summaries, npj Digital Medicine 2025). The dominant *mitigation* is retrieval-augmented generation (RAG): grounding generation in retrieved evidence reduces hallucination substantially, in some clinical settings to near zero (npj Digital Medicine 2025). Yet across this literature three limitations recur. First, **grounding is enforced softly** — by prompting the model to “use only the retrieved passages” — which reduces but cannot *guarantee* that an assertion is evidence-backed, since the model can still contradict or over-reach beyond the retrieved text. Second, systems are **overwhelmingly single-model**; multi-agent LLM verification appears only

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rarely in journal work (Expert Systems with Applications 2025), and its known risk — that agents *agree, confidently, on the same wrong answer* — is unexamined. Third, the **latency and cost** of the reliability mechanism are almost never reported, leaving the practical trade-off unquantified.

This yields a clear **methodological gap**: no empirical study evaluates a *hard, mechanically-verified* evidence gate — one that permits an affirmative or contradictory verdict only when it can quote a verbatim span of the retrieved evidence — as the structural precondition for **multi-agent** claim verification, together with its catch-rate, false-agreement, accuracy, latency and cost trade-offs across model scales and domains. This is the gap the present work addresses. (For technical antecedents, the closest prior art is single-model verbatim gating and soft-grounded debate; both are positioned precisely in §2.)

We close that gap with **Aletheia**, a multi-agent verification pipeline in which agreement is constrained not by a soft, learned, or NLI-scored judgment but by a **hard, mechanical verbatim-substring match**: a verdict may affirm *Supported* or *Contradicted* only by quoting a span the system checks programmatically against the retrieved evidence; failing that check forces *Unverifiable*. This is deliberately narrower than free-form debate — it trades expressiveness for a structural guarantee: two agents cannot both hallucinate agreement on a claim neither can quote support for.

Contributions.

1. **A structural evidence gate** that makes a mechanical verbatim-span match the precondition for multi-agent verifier agreement — to our knowledge the first such construction operating on natural-language evidence (§2, §3).
2. **A deployed system and a seeded, reproducible harness** that reports catch rate, false-agreement, latency and cost against a single-LLM baseline and an ungrounded ablation, all judging the same claims on the same frozen corpus (§3–§5).
3. **An empirical boundary condition**: grounding’s accuracy benefit is inversely related to base-model strength — a large, significant win at 8B, exactly zero at production scale — established across three model scales and two domains (§5, §6).
4. **A demonstrated separation of guarantees**: on the domain built to break verbatim grounding (FEVER), the gate keeps its one promise — 51/51 asserted verdicts are verbatim-backed — while its residual errors are genuine entailment mistakes, not fabrications (§5.6, §6).

2. Related work

The empirical landscape. Journal research on LLM factual reliability clusters into four lines, each mapping to a limitation this work targets. (i) *Factual-accuracy and citation-reliability studies* evaluate whether a model is right, typically in a single domain (Yanagita 2023; GPT-4-vs-Gemini citation reliability, Computers in Biology and Medicine 2024; ChatGPT drug-information accuracy, Khatri 2025) — measuring the problem, not guaranteeing against it. (ii) *Hallucination detection and rate estimation* characterise unsupported output (Massenon 2025; clinical-safety framework, npj Digital Medicine 2025) but detect after the fact rather than prevent. (iii) *Retrieval-augmented mitigation* grounds generation in evidence and consistently reduces hallucination (RAG across 10 LLMs, npj Digital Medicine 2025; long-context and low-resource RAG, JMIR 2025; multilingual RAG verification, Journal of Intelligent Information Systems 2026) — but grounding is prompt-enforced, not mechanically checked. (iv) *Automated verification and misinformation detection* label claims from content, style, or evidence (FacTeR-Check, Knowledge-Based Systems 2022; pretrained-LM detection, Information Processing & Management 2025; and a large body of deep-learning fake-news work, e.g. Kolluri 2022; Lu 2025; Kumar 2025) — but as single-model, black-box classifiers that rarely tie each assertion to a checkable span or report cost. The common thread: reduction without guarantee, single models, and unreported cost.

Direct technical antecedents. Claim-verification benchmarks and systems (FEVER [1], SciFact [2] and its extractive VeriSci baseline, MultiVerS [3]) score a single supervised judgment. Hallucination-detection methods flag unsupported generation without necessarily retrieving evidence: SelfCheckGPT [4] scores consistency across sampled generations; Chain-of-Verification [5] answers self-generated questions from parameters. Multi-agent debate [6] improves factuality through exchange, but a failure-mode literature [a,b,c] shows the exchange can itself produce false consensus. Two lines sit closest and were examined in full text:

hard, mechanically-verified verbatim gates exist (“Show Your Work” [d]) but only for *single-model classification* — no multi-agent structure; and multi-agent debate *for claim verification* is active (“Debating Truth” [e]; TRUST Agents [f]) but grounds evidence *softly* — LLM-judgment moderators, calibrated confidence, or NLI adjudication, never a mechanical string check. Aletheia occupies the intersection these leave open. Consistent with prior evaluation practice (OpenFactCheck [7]), we report an accuracy–latency–cost triad; the catch-rate and false-agreement axes are our additions. This is a *positioning claim*, not a systematic survey; the defensible unit of contribution remains the measured gap to the baseline (§5).

3. System architecture

Aletheia is a LangGraph-orchestrated pipeline of six stages. An **Intake guard** (deterministic prompt-injection scan plus an LLM scope classifier) admits or refuses a query. A **Retriever** performs hybrid semantic + keyword search over a frozen, versioned PostgreSQL/pgvector corpus. A **Generator** proposes an answer decomposed into atomic claims. A per-claim **Verifier** judges each claim against retrieved evidence under the quoted-span discipline. An **Aggregator** collects verdicts and surfaces disagreement explicitly. A non-mutating **Guardrail** attaches a confidence advisory and disclaimer. The LLM client is provider-agnostic (Gemini, Groq, OpenRouter), so one pipeline runs unmodified across all scales and domains compared below.

Two properties are load-bearing. First, **grounding is structural, not aspirational**: pipeline code checks a claimed span against the evidence text (`grounded_against()`) and programmatically downgrades any verdict that cannot produce a matching span — the model is not merely instructed to be honest. Second, **benchmark and live system share one grounding path**: there is no separate “eval mode,” so all reported numbers describe the deployed system.

4. Research questions, hypotheses and method

RQ. Does an evidence-grounded, multi-agent verification pipeline catch measurably more hallucinations than a single LLM, and at what accuracy, latency and cost trade-off, across model scale and domain? **H1:** Aletheia achieves a higher hallucination-catch rate than a single-LLM baseline. **H2:** Requiring quoted-span evidence lowers false-agreement versus an otherwise-identical opinion-only multi-agent arm. **H3:** Reliability gains come at a quantified, defensible latency and cost overhead.

Benchmarks. SciFact [2] (expert scientific claims over 5,183 biomedical abstracts; 15,411 chunks ingested) and FEVER [1] (open-domain claims over Wikipedia, where claims are crowdworker *paraphrases* rather than near-quotations — stress-testing verbatim grounding where it is weakest). **Arms** (all judging the same claim, evidence and model): single-LLM baseline; multi-agent ungrounded ablation (span requirement removed, isolating grounding’s contribution, H2); Aletheia (full grounded verifier). **Sampling:** seeded (seed 7), gold-label-stratified, without replacement; corpus coverage checked before scoring. **Statistics:** paired per-claim tests — exact McNemar for accuracy, percentile-bootstrap 95% CIs (10,000 resamples, fixed seed) for catch-rate and false-agreement deltas. **Fault tolerance:** a provider error on any arm excludes that item from every arm, never fabricating a verdict. **Metrics:** accuracy; catch rate (recall on truly unsupported/false claims); false-agreement rate; latency p50/p95/p99; tokens/query.

5. Results

5.1 Headline benchmark (n=100, current live model)

SciFact · 100 claims · seed 7 · Gemini gemini-3.1-flash-lite-preview · corpus coverage 100.0% · 2026-08-03.

System	Acc	Catch	F-agree	Latency p50/95/99 (s)	Tok/q
Single-LLM baseline	79.0%	93.1%	10.5%	1.15 / 2.89 / 6.55	1291.8
Multi-agent, ungrounded (ablation)	80.0%	94.8%	8.1%	1.37 / 2.62 / 3.11	1364.0
Aletheia (grounded)	79.0%	96.6%	6.1%	1.52 / 3.32 / 5.11	1676.8

Grounded vs baseline (H1): accuracy McNemar $p = 1.000$ (6 discordant); catch delta +3.4 pp, CI [+0.0, +8.9]; false-agree delta -4.5 pp, CI [-12.3, +0.9]. Grounded vs ablation (H2): catch delta +1.7 pp, CI [+0.0, +5.7]; false-agree delta -2.0 pp, CI [-8.1, +1.3].

On a materially stronger base model, grounding’s accuracy edge disappears (79.0% vs 79.0%, exactly tied); catch-rate and false-agreement point the right direction but both intervals touch or cross zero at $n=100$. The grounded arm costs ~30% more tokens at higher latency (H3), for a benefit present in direction but not statistically confirmed at this n . §5.3 identifies the mechanism.

5.2 Weaker-model comparison ($n=100$, 8B)

SciFact · 100 claims · seed 7 · Groq llama-3.1-8b-instant · corpus coverage 100.0% · 2026-07-19.

System	Acc	Catch	F-agree	Latency p50/95/99 (s)	Tok/q
Single-LLM baseline	60.0%	60.3%	37.7%	0.30 / 8.93 / 12.95	1388.0
Multi-agent, ungrounded (ablation)	65.0%	65.5%	35.7%	14.06 / 19.00 / 21.66	1473.2
Aletheia (grounded)	69.0%	82.8%	23.8%	0.41 / 0.57 / 1.11	1675.6

Catch 82.8% vs 60.3%, delta +22.4 pp, CI [+12.1, +33.3] (excludes zero); accuracy 69.0% vs 60.0%, +9.0 pp (McNemar $p = 0.163$); false-agree 23.8% vs 37.7%, delta -13.9 pp, CI [-23.9, -5.0].

On this weaker model the primary thesis metric holds under paired significance and by a wide margin, establishing that grounding’s effect is real and large when the base model is weak enough to need the correction.

5.3 Error decomposition

Outcome	8B (§5.2)	Gemini (§5.1)
Correct	69	79
Retrieval miss	—	1
Verifier abstention (evidence present, no span quoted)	14/63	9/63
Wrong direction	6	4
False grounding (gold Unverifiable, verdict asserts)	10/37	7/37

Retrieval is not the bottleneck at either scale (at most 1/100 gold passages missed); errors are overwhelmingly verifier decisions. The stronger model needs the strict single-span discipline less often on both sides — false-grounding 10/37->7/37, abstention 14/63->9/63 — yet this does not convert into an accuracy edge, because the ungrounded baseline is already strong enough to get most claims right unaided.

Caveat on false-grounding. SciFact’s **NotEnoughInfo** label is defined against its own annotated evidence set, while Aletheia retrieves from the full corpus — so a verdict counted as false grounding sometimes has genuine, uncited support. Both figures are upper bounds on verifier error, not clean counts of hallucination; separating them needs manual adjudication (§7).

5.4 Cross-model robustness

$n=19-30$; every delta below is not statistically significant — read for direction and mechanism; §5.1 confirms the mechanism at a powered n .

Base model	n	Baseline acc	Grounded acc	D-acc	D-catch	D-fagree
8B	30	56.7%	66.7%	+10.0	+17.6	-11.9
70B	30	80.0%	70.0%	-10.0	+5.9	-6.0
550B-class	19	89.5%	73.7%	-15.8	+0.0	+0.0

Catch-rate and false-agreement advantages hold at every scale (at 550B the baseline is at ceiling). But the **accuracy effect flips sign with base-model strength**: +10 pp at 8B, -10 to -16 pp at 70B/550B, and 0 pp at Gemini scale (§5.1). The residual error shifts to over-abstention: a strong model that would answer correctly is forced to *Unverifiable* when it cannot isolate one verbatim span.

5.5 A targeted verifier improvement

A two-sided **span-sufficiency test** (assert only when the span alone decides the claim; do not retreat when it plainly does; treat merely-topical spans as *Unverifiable*) raised 8B accuracy 53.3%→66.7% (+13.4 pp, seed 13) with catch-rate unchanged. Re-run at 70B, it halves false-grounding (4/11→2/11) but raises abstention (5/19→7/19), so accuracy slips 70.0%→66.7% while catch reaches 100%. The change **sharpens** §5.4’s trade-off rather than escaping it. §5.1 already uses this improved prompt.

5.6 Second domain: FEVER

FEVER · 100 claims · seed 7 · Groq llama-3.1-8b-instant · corpus coverage 99.0% · 2026-07-25.

System	Acc	Catch	F-agree
Single-LLM baseline	80.0%	84.8%	23.8%
Multi-agent, ungrounded (ablation)	85.0%	95.5%	9.1%
Aletheia (grounded verifier)	77.0%	93.9%	13.8%

H1 (vs baseline): accuracy McNemar $p = 0.648$; catch delta +9.1 pp, CI [+1.5, +17.7]; false-agree delta -10.0 pp, CI [-22.9, +2.0]. H2 (vs ablation): accuracy McNemar $p = 0.008$ (grounded disfavour); false-agree delta +4.7 pp, CI [+0.3, +12.6].

The corpus lever: 56.0% → 77.0% accuracy (+21 pp), McNemar $p = 4.9\text{e-}5$. An earlier run on the identical claims/model scored 56.0% accuracy. The only change is the corpus: FEVER’s wiki text ships in Penn-Treebank tokenization (-LRB-/RRB-, spaced punctuation), which an 8B model paraphrases away while copying an otherwise-correct span — the guard then rejected a *correct* quote and forced abstention. Folding the markup to plain text fixed 24 grounded verdicts and broke 3: the earlier 56% was overwhelmingly a corpus-markup artifact, not a reasoning ceiling.

The guarantee held. Of the 51 asserted verdicts, **51/51 (100%) are backed by a span appearing verbatim in the retrieved evidence**, re-verified from traces. The 5 wrong assertions are genuine *entailment* errors — the reader quoted faithfully and inferred wrongly — not fabrications and not guard failures. After the fix the grounded arm (77.0%) trails the ungrounded ablation (85.0%): strict verbatim grounding costs ~8 pp of raw accuracy on paraphrase claims (H2 significant, reported without softening), while still beating the baseline on catch rate (H1, CI excludes zero).

6. Discussion

What the evidence supports. Across three model scales and two domains, the structural gate’s *safety* direction is consistent: catch rate rises and false-agreement falls versus the single-LLM baseline in every configuration, significantly so where the base model is weak (§5.2) and on FEVER-vs-baseline catch rate (§5.6). H1 is therefore supported in direction throughout and significantly at the scales with room to improve; H2 is directionally supported but not consistently significant at n=100. Most robust of all is the *mechanical guarantee*: 51/51 verbatim-backed assertions on the domain designed to break it. This is the result a reviewer can check independently from the traces, and it does not depend on a sample size.

The central scientific finding is a boundary condition, not a leaderboard win. Grounding’s contribution to *accuracy* is inversely related to base-model strength (§5.1, §5.4): a strong model already answers most claims correctly, so the span discipline has little error left to correct and instead occasionally forces a correct answer into abstention. This reframes when structural grounding is worth its cost. It is a net accuracy win for weaker or cheaper models; on strong models its value is not accuracy but *auditability* — a guarantee that every assertion is quote-backed, at ~30% token overhead.

A conceptual separation the field should adopt. Aletheia guarantees *no unsupported assertion*, not *correct entailment*. The FEVER errors make this concrete: the model quoted the evidence exactly and still reasoned wrongly. Conflating the two guarantees overstates what any verbatim gate can promise; separating them clarifies that the gate eliminates fabrication while leaving reading comprehension as a distinct, remaining problem.

Practical implication. For safety-critical or low-cost deployments — where a confidently wrong shared verdict is costlier than an honest abstention — a hard evidence gate is the appropriate default. For maximum accuracy on strong models with lenient auditing needs, a softer check may dominate; quantifying that crossover is the natural next study (§7).

7. Limitations and future work

- **No soft-grounding baseline yet.** The reported ablation is hard-gate vs. no-check; a third NLI-threshold arm (MiniCheck/AlignScore style) is needed to isolate hard-gate vs. soft-check — the comparison closest to §2’s related work and the first a reviewer will request.
- **No leaderboard comparison.** All arms are internal; a comparison against MultiVerS (SciFact) and a FEVER/AVerITeC entry is absent and flagged, not omitted silently.
- **Small-n exploratory sections** (§5.4, §5.5 re-checks): every delta there is insignificant and read only for mechanism.
- **Single seeded repeat at n=100; --repeats N** would tighten every CI.
- **False-grounding is an upper bound** (§5.3) pending manual adjudication of uncited-but-genuine support.
- **Threats to validity:** possible benchmark contamination (mitigated by reporting relative gaps), prompt sensitivity (controlled by fixing prompts within a comparison), and retriever ceiling (measured directly, near-zero here but corpus-specific).

8. Conclusion

We presented Aletheia, a multi-agent claim-verification pipeline that makes a hard, mechanical verbatim-substring match the structural precondition for verifier agreement, and evaluated it as a deployed system with a seeded, reproducible harness. The gate is a mechanically verifiable defence against false agreement: its catch-rate and false-agreement advantages over a single-LLM baseline hold in direction across three model scales and two domains, significantly where the base model is weak, and its one promise — never assert without a verbatim quote — is kept exactly (51/51) even on the domain built to break it. Its effect on aggregate accuracy is not a free lunch but a scale- and domain-dependent trade-off: a real gain for weaker models, an auditability guarantee rather than an accuracy gain for strong ones. Reporting where structural grounding helps, where it costs, and why — including the results that do not flatter the thesis — is itself the contribution.

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Empirical journal literature reviewed (representative; the full 35-paper review is provided as a supplementary literature table). Author surnames to be completed from each DOI before submission:

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- a. “Talk Isn’t Always Cheap.” arXiv:2509.05396.
- b. “Peacemaker or Troublemaker.” arXiv:2509.23055.
- c. “The Deliberative Illusion.” arXiv:2606.03032.
- d. “Show Your Work.” medRxiv 2026.03.03.26346690 / *Cureus* 2026.
- e. “Debating Truth.” arXiv:2507.19090 (WWW 2026).
- f. “TRUST Agents.” arXiv:2604.12184.

All numeric results are traceable to committed seeded runs in the project’s evaluation record; no value is projected. Two experiments named in §7 (soft-grounding baseline arm; leaderboard comparison) remain before submission to a top-tier venue.